LAB ASSIGNMENT 2

NAME – VANSHIKA DIXIT

ROLL NUMBER - 2401010010

Objective:

• To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

Instructions:

1. **Open your preferred Database Management System (DBMS) tool.**
2. **Create the required tables: CLASS, EMPLOYEES, and Client_master.**
3. **Insert the given/suitable records into each table.**
4. **Execute the SQL scripts and queries step by step.**
5. **Observe the output carefully after each operation, especially after ROLLBACK and UPDATE commands.**

Table Structures:

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

EMPLOYEES

Employee_Id (NUMBER), First_Name (VARCHAR), Last_Name (VARCHAR), Email (VARCHAR), Phone_Number (VARCHAR), Hire_Date (DATE), Job_Id (VARCHAR), Salary (NUMBER), Commission_Pct (NUMBER), Manager_Id (NUMBER), and Department_Id (NUMBER).

DDL Questions (Data Definition Language):

1. **How do you create a table named CLASS with appropriate data types?**

1. **How do you create a table named EMPLOYEES with the specified attributes?**
2. **How do you create a table named Client_master with the specified attributes?**
3. **What SQL statement would you use to rename the table Client_master to Client12?**
4. **How do you display the bal_due column heading as "BALANCE"?**

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

- **INSERT INTO class VALUES (5,'Rahul');**
- **COMMIT;**
- **UPDATE class SET name = 'Abhijeet' WHERE id='5';**
- **SAVEPOINT A;**
- **INSERT INTO class VALUES (6,'Chris');**
- **SAVEPOINT B;**
- **INSERT INTO class VALUES (7,'Bravo');**
- **SAVEPOINT C;**
- **SELECT * FROM class;**
- **ROLLBACK TO B;**
- **SELECT * FROM class;**
- **ROLLBACK TO A;**

Part B: Retrieving Records (EMPLOYEES Table)

1. **Find out the employee id, names, and salaries of all the employees.**
2. **List out the employees who work under manager 100.**
3. **Find the names of the employees who have a salary greater than or equal to 4800.**
4. **List out the employees whose last name is 'AUSTIN'.**
5. **Find the names of the employees who work in departments 60, 70, and 80.**
6. **Display the unique Manager_Id.**

Part C: Updating Records (Client_master Table)

1. **Insert five records into the Client_master table.**
2. **Find the names of clients whose bal_due > 5000.**
3. **Change the bal_due of ClientNO "C123" to Rs. 5100.**

```
CREATE DATABASE DBMS_Vanshika_Dixit;
USE DBMS_Vanshika_Dixit;

CREATE TABLE CLASS (
   id INT,
   name VARCHAR(50)
);

INSERT INTO CLASS VALUES (1, 'Vanshika');
INSERT INTO CLASS VALUES (2, 'Roshani');
INSERT INTO CLASS VALUES (4, 'Bhoomi');

CREATE TABLE EMPLOYEES (
   Employee_Id INT PRIMARY KEY,
   First_Name VARCHAR(50),
   Last_Name VARCHAR(50),
   Email VARCHAR(100),
   Phone_Number VARCHAR(20),
   Hire_Date DATE,
   Job_Id VARCHAR(50),
   Salary DECIMAL(10,2),
   Commission_Pct DECIMAL(4,2),
   Manager_Id INT,
   Department_Id INT
);
```

```
CREATE TABLE Client_master (
   ClientNo VARCHAR(10) PRIMARY KEY,
   Name VARCHAR(50),
   Address VARCHAR(100),
   City VARCHAR(50),
   State VARCHAR(50),
   Pincode INT,
   Bal_due DECIMAL(10,2)
);

RENAME TABLE Client_master TO Client12;

SELECT Bal_due AS "BALANCE" FROM Client12;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	BALANCE

```
INSERT INTO CLASS VALUES (5,'Rahul');
COMMIT;

UPDATE CLASS SET name='Abhijeet' WHERE id=5;
SAVEPOINT A;

INSERT INTO CLASS VALUES (6,'Chris');
SAVEPOINT B;

INSERT INTO CLASS VALUES (7,'Bravo');
SAVEPOINT C;

SELECT * FROM CLASS;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

id	name
1	Vanshika
2	Roshani
4	Bhoomi
6	Chris
7	Bravo

ROLLBACK TO B;

SELECT * FROM CLASS;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name
1	Vanshika
2	Roshani
4	Bhoomi
5	Abhijeet
6	Chris
NULL	NULL

ROLLBACK TO A;

SELECT * FROM CLASS;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name
1	Vanshika
2	Roshani
4	Bhoomi
5	Abhijeet
NULL	NULL

```
CREATE TABLE EMPLOYEES (
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Email VARCHAR(100),
    Phone_Number VARCHAR(20),
    Hire_Date DATE,
```

```
    Job_Id VARCHAR(50),
    Salary DECIMAL(10,2),
    Commission_Pct DECIMAL(4,2),
    Manager_Id INT,
    Department_Id INT
);

INSERT INTO EMPLOYEES VALUES
(101,'John','AUSTIN','john@gmail.com','9876543210','2020-01-10','IT_PROG',5000,NULL,100,60),
(102,'Sara','James','sara@gmail.com','9876543211','2019-03-15','HR_REP',4500,NULL,101,70),
(103,'Mike','Taylor','mike@gmail.com','9876543212','2021-06-20','SA_REP',6000,0.10,100,80),
(104,'Anna','AUSTIN','anna@gmail.com','9876543213','2018-07-25','IT_PROG',4800,NULL,102,60),
(105,'David','Lee','david@gmail.com','9876543214','2022-02-11','SA_REP',7000,0.15,100,80);

SELECT Employee_Id, First_Name, Last_Name, Salary FROM EMPLOYEES;
```

Result Grid | Filter Rows: | Edit: | Export/

Employee_Id	First_Name	Last_Name	Salary
101	John	AUSTIN	5000.00
102	Sara	James	4500.00
103	Mike	Taylor	6000.00
104	Anna	AUSTIN	4800.00
105	David	Lee	7000.00
NULL	NULL	NULL	NULL

```
SELECT * FROM EMPLOYEES WHERE Manager_Id = 100;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	John	AUSTIN	john@gmail.com	9876543210	2020-01-10	IT_PROG	5000.00	NULL	100	60
103	Mike	Taylor	mike@gmail.com	9876543212	2021-06-20	SA_REP	6000.00	0.10	100	80
105	David	Lee	david@gmail.com	9876543214	2022-02-11	SA_REP	7000.00	0.15	100	80
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

SELECT First_Name, Last_Name FROM EMPLOYEES WHERE Salary >= 4800;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First_Name	Last_Name
John	AUSTIN
Mike	Taylor
Anna	AUSTIN
David	Lee

SELECT * FROM EMPLOYEES WHERE Last_Name = 'AUSTIN';

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	John	AUSTIN	john@gmail.com	9876543210	2020-01-10	IT_PROG	5000.00	NULL	100	60
104	Anna	AUSTIN	anna@gmail.com	9876543213	2018-07-25	IT_PROG	4800.00	NULL	102	60
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

SELECT First_Name, Last_Name FROM EMPLOYEES

WHERE Department_Id IN (60,70,80);

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First_Name	Last_Name
John	AUSTIN
Sara	James
Mike	Taylor
Anna	AUSTIN
David	Lee

SELECT DISTINCT Manager_Id FROM EMPLOYEES;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Manager_Id
100
101
102

CREATE TABLE Client_master (

ClientNo VARCHAR(10) PRIMARY KEY,

Name VARCHAR(50),

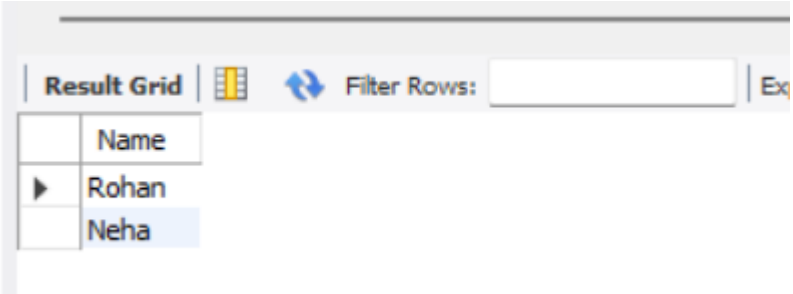
Address VARCHAR(100),

```
    City VARCHAR(50),
    State VARCHAR(50),
    Pincode INT,
    Bal_due DECIMAL(10,2)
);

RENAME TABLE Client_master TO Client12;

INSERT INTO Client12 VALUES
('C101','Rohan','MG Road','Mumbai','Maharashtra',400001,6000),
('C102','Priya','Brigade Road','Bangalore','Karnataka',560001,3000),
('C123','Amit','Park Street','Kolkata','West Bengal',700016,4500),
('C104','Neha','Banjara Hills','Hyderabad','Telangana',500034,8000),
('C105','Raj','Connaught Place','Delhi','Delhi',110001,2000);

SELECT Name FROM Client12
WHERE Bal_due > 5000;
```




```
UPDATE Client12
SET Bal_due = 5100
WHERE ClientNo = 'C123';

SELECT * FROM Client12 WHERE ClientNo='C123';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ClientNo	Name	Address	City	State	Pincode	Bal_due
C123	Amit	Park Street	Kolkata	West Bengal	700016	5100.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL